MIL-B-52523A(ME)
30 April 1984
SUPERSEDING
MIL-B-52523(ME)
1 September 1966

MILITARY SPECIFICATION

BLOCKS, TACKLE, WIRE ROPE, ORDINARY, 10-INCH SHEAVE

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ordinary, wire rope tackle blocks having 10-inch sheaves.

1.2 Classification. Blocks shall be the following types, as specified (see 6.2):

Type I	- Fall (weighted), single sheave, diamond steel shell, for 3/8-inch rope.
Type II	- Regular, double sheave, oval steel shell, for 3/8-inch rope.
Type III	- Regular, single sheave, oval steel shell, for 1/2-inch rope.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-3865	- Blocks, Rope, Tackle: Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15024/6	- Plates, Identification, Equipment.
MIL-G-20241	- Material, Wool Felt Impregnated, Adhesive, Ressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13208E5017	- Block, Tackle, Wire Rope, Double, 10-Inch Sheaves
TA13208E5018	- Block, Tackle, Wire Rope, Single, 10-Inch Sheave Single, 10-Inch Sheave

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TA13208E5077 - Block, Tackle, Wire Rope, Fall (Weighted), Single, 10-Inch Sheave

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The blocks shall be as shown on the applicable drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the blocks. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The blocks shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and/or operating environment to which the item may be exposed.

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3.3.1.1 Dissimilar Metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or his/her designated representative.

3.4 Threads. Threads shall conform to FED-STD-H28

3.5 Performance.

3.5.1 Safe working load. The blocks shall function as specified herein under the safe working loads (SWL) shown in table I, without load, the sheaves shall turn freely by hand. While under their relative shown loads, the blocks shall rotate through 360 degrees upon their respective fitting without binding.

TABLE I. Safe working load (SWL).

Block type	Sheave OD	No. of sheaves	Rope diameter	Block fitting	SWL
	Inches		Inch		Pounds
I	10	1	3/8	Safety swivel hook	8,000
II	10	2	3/8	Swivel shackle	6,000
III	10	1	1/2	Swivel shackle	3,000

3.5.2 Proof load. The blocks shall withstand proof loads twice their respective safe working loads shown in table I without evidence of distortion, permanent set, deformation, binding, twisting, or freezing of any part.

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3.5.3 Ultimate strength. Blocks tested to their ultimate strength shall fail by either of the following occurences, whichever is applicable, prior to failure of any of the other components of the blocks:

a. Dropping of the load by opening and straightening of the hook.

b. Rupture of the shackle.

3.6 Lubrication. Blocks shall be lubricated with greases conforming to MIL-G-10924. The blocks shall be assembled, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not compatible with other greases. Each block shall be tagged in a conspicuous place to indicate which military grease has been used.

3.7 Identification marking. The blocks shall be identified in accordance with MIL-STD-130. The marking shall be applied to the blocks on plates conforming to MIL-P-15024/6, type H. In addition, the safe working load, in pounds, of each block shall be permanently stamped on the blocks.

3.8 Treatment and painting. The portions of the blocks normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704.

3.9 Type I, fall block. The type I block shall be as shown on Top Assembly TA13208E5077 and as specified herein.

3.10 Type II, regular block. The type II block shall be as shown on TA13208E5017 and as specified herein.

3.11 Type III, regular block. The type III block shall be as shown on TA13208E5018 and as specified herein.

3.12 Workmanship. All parts, components, and assemblies of the blocks, including forgings, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free of rust, scale, scrapes, burrs, dirt, pits, flux, and other harmful extraneous material. All edges of the blocks shall be rounded or chamfered, and surfaces shall be smooth to the touch.

3.12.1 Welding. The surfaces of parts to be welded shall be free of rust, scale, paint, grease, and other foreign material. Welds shall transmit ultimate strength stresses between the parts connected without evidence of failure.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Blocks failing a performance test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. parts not complying with such requirements shall be rejected.

4.1.3 Parts and components. Parts and components shall be inspected in accordance with the drawings and Quality assurance provisions (QAP) listed on TA13208E5017, TA13208E5018, and TA13208E5077. Evidence that any part or component does not comply with the applicable drawings shall be cause for rejection of that part or component.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article block shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article block shall be tested as specified in 4.5.2.1 through 4.5 2.3. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of either test shall be cause for rejection.

4.5. Inspection procedure.

4.5.1 Examination. The blocks shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and/or operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Dimensions not as shown on the drawings.
106. Threads not as specified.
107. Sheaves do not turn freely by hand.
108. Workmanship not as specified.
109. Welding not as specified.

Minor

201. Lubrication not as specified.
202. Identification marking not as specified.
203. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Safe working load. Reeve the block with the size wire rope shown in table I, and suspend the wire rope between adequately supported posts. Attach a load, equal to the SWL shown in table I, to the hook or shackle of the block, and run the block along the wire rope for a distance of 25 feet in each direction. Rotate the load about the hook or shackle fitting through 360 degrees. Nonconformance to 3.5.1 shall constitute failure of this test.

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4.5.2.2 Proof load test. Attach the block to the fixed head of a tension testing machine or device by means of wire rope or round steel bar loops passed around the sheave or sheaves of the block. Then attach the hook or shackle of the block to the moving head of the machine or the moving load by suitable links or connections. Increase the load to the block to twice the safe working load shown in table I. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3 Ultimate strength test. Attach the block to a testing device as specified in 4.5.2.2. Apply a gradually increasing load until the hook begins to open and straighten, or the shackle begins to rupture. Then examine the block and its components for signs of deformation. Deformation of any component other than the hook or shackle shall constitute failure of this test.

4.6 Inspection of packaging. The packaging of the blocks shall be examined to determine compliance with the requirements of section 5 of this specification and the document referenced therein.

5. PACKAGING

5.1 Preservation, packing, palletization, and marking. Preservation, packing, palletization, and marking shall be in accordance with the requirements of MIL-B-3865 except that unless otherwise specified (see 6.2), type I blocks shall be shipped loose, unpacked. Preservation shall be level A or commercial (see 6.2). Packing shall be level A, level B, or commercial (see 6.2). Palletization shall be level A, level B, or commercial, as applicable.

6. NOTES

6.1 Intended use. The tackle blocks covered by this specification are used in the erection and operation of Army medium aerial cableway sets.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of block required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2, 4.3, and 6.3).

d. Degree of preservation and degree of packing required (see 5.1).

e. When type I blocks shall be packed (see 5.1).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52523A(ME)	Blocks, Tackle, Wire Rope, Ordinary, 10-Inch Sheave

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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